1) FLOPS

- Conv FLOPS = $\frac{2 \cdot K \cdot K \cdot C_{in} \cdot S^2 \cdot C_{out} \cdot B}{stride^2}$

, where K = Kernel size,
C_{in} = input channels
C_{out} = output channels
S = image size, B - batch size

- ReLU - 0 FLOPS, because the block has 0 MAC operations
- MaxPool - 0 FLOPS, same logic as
- Linear layer FLOPS - $2 \cdot B \cdot in_features \cdot out_features$
- Global/Avg Pool - $B \cdot C \cdot S \cdot S$, C - channels
S - size of input image
B - batch size

2) Memory (S, B)

Memory (S, B) = sum (weights, activations)
Conv_weights = $C_{in} \cdot K \cdot K \cdot C_{out}$
ReLU_weights, Max/Avg Pool = 0
Linear_weights = in_features * out_features

3) Bytes moved = sum(weights_size + input_size + output_size) -
- equation for each layer of network.

4) Latency = $\max\left(\frac{\text{FLOPS}}{\text{peak FLOPS}}, \frac{\text{Bytes moved}}{\text{memory bandwidth}}\right)$

5) Energy = $\alpha \cdot \text{FLOPS} + \beta \cdot \text{Bytes Moved}$

need to calibrate

FLOPS calculation

$$FLOPS(S,B) = \frac{2 \cdot 7^2 \cdot 3 \cdot S^2 \cdot 32 \cdot B}{4} + \frac{2 \cdot 5^2 \cdot 32 \cdot \left(\frac{S}{2}\right)^2 \cdot 64 \cdot B}{1} +$$

$$+ \frac{2 \cdot 3^2 \cdot 64 \cdot \left(\frac{S}{4}\right)^2 \cdot 128 \cdot B}{4} + \frac{2 \cdot 1^2 \cdot 128 \cdot \left(\frac{S}{8}\right)^2 \cdot 256 \cdot B}{1} +$$

$$+ \frac{2 \cdot 3^2 \cdot 256^2 \cdot \left(\frac{S}{8}\right)^2 \cdot B}{4} + \frac{2 \cdot 1^2 \cdot 256 \cdot \left(\frac{S}{16}\right)^2 \cdot 512 \cdot B}{1} +$$

$$+ B \cdot 512 \cdot \left(\frac{S}{16}\right)^2 + 2 \cdot B \cdot 512 \cdot 256 + 2 \cdot B \cdot 256 \cdot 100 =$$

$$= S^2 \cdot B (2352 + 6400 + 2304 + 1024 + 4608 + 1024 +$$

$$+ 27) + B \cdot (262144 + 51200) = \frac{17714}{26} \cdot S^2 \cdot B +$$

$$+ 313344 \cdot B$$

Memory calculation

$$Memory(S,B) = 7 \cdot 7 \cdot 3 \cdot 32 + 5 \cdot 5 \cdot 32 \cdot 64 + 3 \cdot 3 \cdot 64 \cdot 128 +$$

$$+ 128 \cdot 256 + 3 \cdot 3 \cdot 256 \cdot 256 + 256 \cdot 512 + 512 \cdot 256 + 256 \cdot 100 +$$

$$+ B \cdot 3 \cdot S^2 + B \cdot 32 \cdot \left(\frac{S}{2}\right)^2 + B \cdot 32 \left(\frac{S}{4}\right)^2 + B \cdot 64 \cdot \left(\frac{S}{4}\right)^2 +$$

$$+ B \cdot 128 \cdot \left(\frac{S}{8}\right)^2 + B \cdot 256 \left(\frac{S}{8}\right)^2 + B \cdot 256 \left(\frac{S}{16}\right)^2 + B \cdot 512 \left(\frac{S}{16}\right)^2 +$$

$$+ B \cdot 512 + B \cdot 256 + B \cdot 100 = 1010400 + B \cdot S^2 \cdot$$

$$\cdot \frac{1039968}{26} + B(512 + 256 + 100) = 4 \cdot (1010400 + B \cdot S^2 \cdot 26$$

$$+ 868 \cdot B)$$

Bytes moved

1039968

Bytes moved $(\leq, B) = \cancel{A \cdot A \cdot 3 \cdot 32}(\cancel{1010400} +$

$\cancel{+B \cdot 3 \cdot 32} + B \cdot 5^2 \cdot 49 + B \cdot 1836) \cdot 4$